



Algorithmic Strategy Development on Multi-Feature Time Series

Idea Summary

1 Introduction

In this project, we developed systematic intraday trading strategies for two anonymized datasets, EBX and EBY, each featuring a core price series and hundreds of masked indicators. Our first approach identified local extremes using normalized second derivatives of band-based features and non-linear price-volume transformations. We then implemented "Adaptive Sigma Flow" utilizing standard and weighted Z-scores alongside Tick Sign Sum (TSS) for momentum validation. The final architecture employs a Reinforcement Learning pipeline based on Proximal Policy Optimization (PPO) to learn an optimal policy that maximizes risk-adjusted returns across diverse intraday regimes.

2 Experimented Approaches

2.1 Strategy 1

Our first approach focuses on identifying local price extremes by leveraging acceleration in band-based features and non-linear transformations of price-volume indicators. This strategy identifies structural exhaustion points within 1-second interval data.

2.1.1 Band-Based Feature Acceleration (BB5_T3)

To capture local reversals, we utilize the second derivative of band-based (BB) features to measure curvature. The process involves:

- **Raw Input:** Starting with the raw feature BB_{raw} .
- **Curvature Calculation:** We compute the second derivative to capture acceleration:

$$\ddot{B}B = BB_t - 2BB_{t-50} + BB_{t-100} \quad (1)$$

- **Normalization:** The signal is standardized to produce the normalized curvature metric.

$$\ddot{B}B_{norm} = \ddot{B}B / \sigma_{BB}. \quad (2)$$

2.1.2 Price-Volume Transformation (PV3_B4_T1)

We map liquidity flows into price-space to identify structural boundaries:

- **Smoothing:** PV_{raw} is smoothed via $SMA(200)$ to yield PV_{smooth} .
- **Scaling:** We scale the volume indicator between 0 and 1:

$$\hat{P}V = \frac{PV - \min(PV)}{\max(PV) - \min(PV)} \quad (3)$$

- **Price Mapping:** The indicator is mapped to the price range to create PV_{scaled} :

$$PV_{scaled} = \min(P) + (\hat{P}V \times Range) \quad (4)$$

Where $Range = \max(P) - \min(P)$.

2.1.3 Trading Signals

Signals are triggered when price interacts with the PV_{scaled} boundaries under specific curvature and trend conditions:

- **BUY Signal:** $PV_{scaled} < Price$, $\ddot{BB}_{norm} \geq 2$, and $BB5_T5_t > BB5_T5_{t-20}$.
- **SELL Signal:** $PV_{scaled} > Price$, $\ddot{BB}_{norm} \leq -2$, and $BB5_T5_t < BB5_T5_{t-20}$.

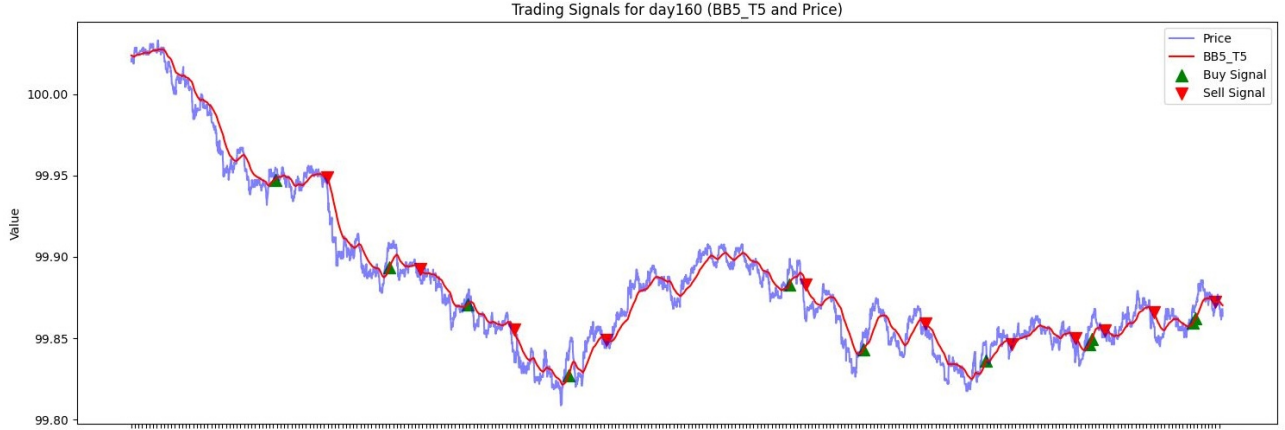


Figure 1: Visualization of Trading Signals on Intraday Data.

Table 1: Average Annualized Performance Summary

Ticker	Ann.PnL(%)	Calmar	Max DD	Avg.Trades/day	Win Rate
EBX(480 days)	-6.63	-0.72	9.12	10.77	73.33
EBY(220 days)	-4.16	-0.66	6.27	10.85	72.8

Although the strategy identifies accurate entry and exit points, it fails to generate a positive net PnL. The individual trade profits are too marginal to overcome transaction costs, resulting in a negative net return once execution fees are accounted for.

2.2 Strategy 2

2.2.1 Intuition

It is designed to capture alpha by identifying statistical overextensions in asset prices while neutralizing low-volatility noise. The entry logic specifically targets Low-Volatility Extremes, where the price reaches a statistical boundary after a period of stable trend. These entries are only executed after confirming the immediate order-flow direction to ensure the trade aligns with a deceleration in opposing momentum.

Conversely, the exit logic is engineered for High-Volatility Extremes. By utilizing a Volatility-Weighted Score, the strategy identifies price movements that occur in very short, highly volatile spans. Profits are taken when the second-derivative Curvature (C_t) confirms that the momentum of these volatile anomalies has mathematically decelerated, ensuring exits occur at the structural inflection point of trend exhaustion.

2.2.2 Mathematical Core: The Signal Engine

The strategy calculates three distinct variation and flow metrics to identify entries and exits:

- **Standard Z-Score (Z_t):** A normalized measure used for entry signals:

$$Z_t = \frac{P_t - \mu_{1500}}{\sigma_{100}} \quad (5)$$

Where μ_{1500} is the 1500-tick SMA and σ_{100} is the 100-tick Rolling Standard Deviation.

- **Tick Sign Sum (TSS):** A flow metric representing the rolling order flow over 100 ticks:

$$TSS = \sum_{i=0}^{100} \frac{P_t - P_{t-1}}{|P_t - P_{t-1}|} \quad (6)$$

Positive values indicate buying pressure, while negative values indicate selling pressure.

- **Volatility-Weighted Score (Z_w):** An amplified measure for exit signals:

$$Z_w = (P_t - \mu_{1500}) \times \sigma_{100} \quad (7)$$

This rewards high-volatility deviations to identify “blow-off” tops or panic bottoms.

- **Signal Line (S_t):** The primary smoothed indicator:

$$S_t = SMA(Z_t, 400) \quad (8)$$

- **Curvature (C_t):** The second derivative of the signal line used to detect momentum deceleration and structural inflection points.

$$C_t = S_t - 2S_{t-150} + S_{t-300} \quad (9)$$

2.2.3 Regime Adaptation: Stabilized Thresholds

To prevent noise triggers during low-volatility regimes, the strategy employs dynamic bands with an integrated Safety Floor (k) for both signal types.

A. Standard Z-Score (Z_s) Thresholds

- **Safety Floor (k):** $k = \max(1, \text{StdDev}(Z_s, 2000))$
- **Upper Band (A^{upper}):** $\text{SMA}(Z_s, 2000) + 2k$
- **Lower Band (A^{lower}):** $\text{SMA}(Z_s, 2000) - 2k$

B. Weighted Z-Score (Z_w) Thresholds

- **Safety Floor (k):** $k = \text{StdDev}(Z_w, 4000)$
- **Upper Band (B^{upper}):** $\text{SMA}(Z_w, 4000) + 2k$
- **Lower Band (B^{lower}):** $\text{SMA}(Z_w, 4000) - 2k$

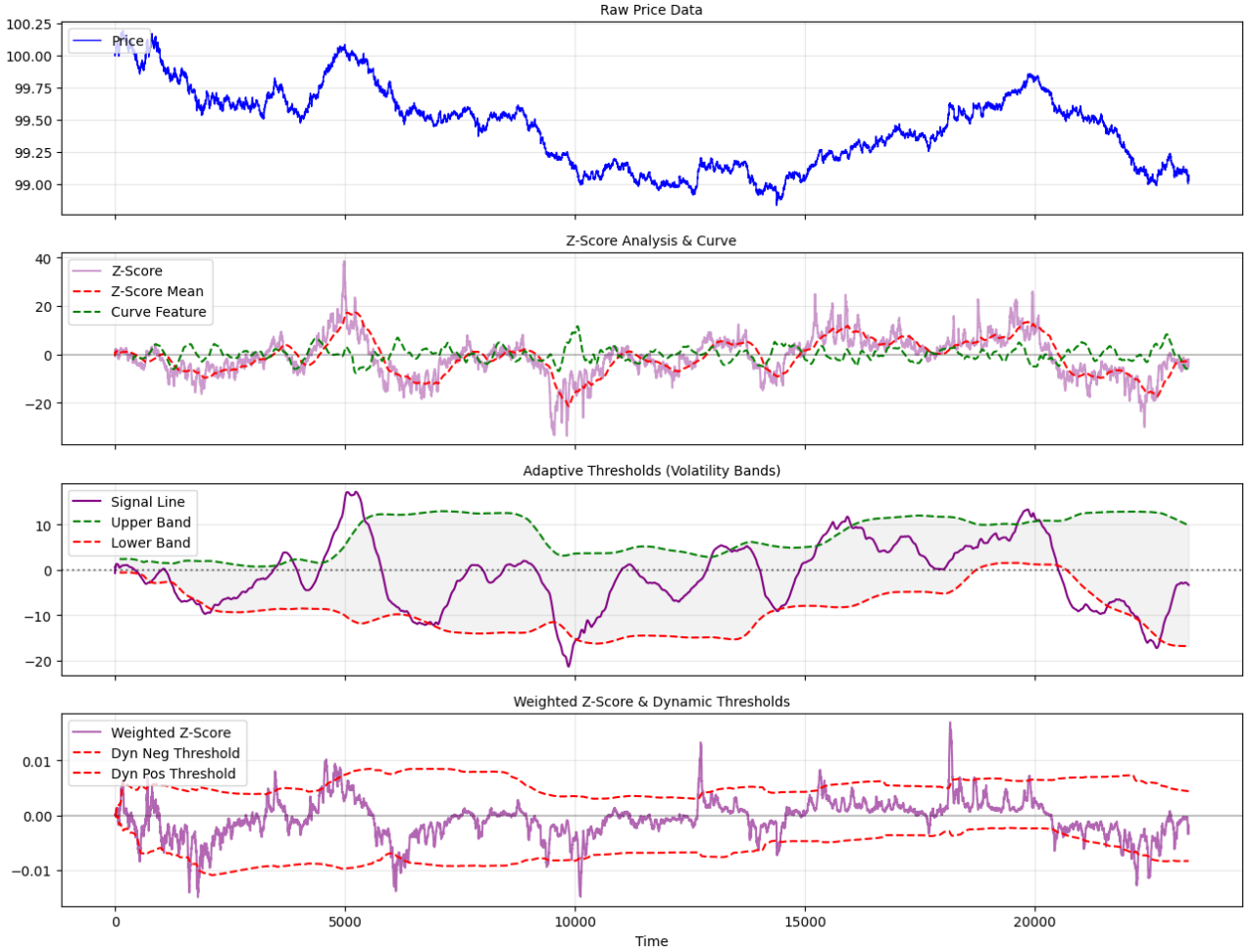


Figure 2: Dynamic Threshold Boundaries for Standard and Weighted Z-Scores

2.2.4 Signal Generation Logic

A. Entry Logic (Counter-Trend with Flow Confirmation)

Trades are initiated when the normalized price breaches the established bands and is confirmed by order flow momentum (TSS).

- **Long Entry:** $(S_{t-250} > A_{t-250}^{\text{lower}})$ AND $(S_t \leq A_t^{\text{lower}})$ AND $(\text{TSS} \leq -50)$
- **Short Entry:** $(S_{t-250} < A_{t-250}^{\text{upper}})$ AND $(S_t \geq A_t^{\text{upper}})$ AND $(\text{TSS} \geq 50)$

B. Exit Logic (Momentum Exhaustion)

Profit taking occurs when signal acceleration (curvature) confirms momentum deceleration at extreme price levels.

- **Long Exit:** $(C_t < -1)$ AND $(Z_w > B^{\text{upper}})$
- **Short Exit:** $(C_t > 1)$ AND $(Z_w < B^{\text{lower}})$

Table 2: Average Annualized Performance Summary

Ticker	Ann.PnL(%)	Sharpe	Calmar	Max DD	Avg.Trades/day	Win Rate
EBX(480 days)	19.07	2.02	4.48	4.25	1.81	71.05
EBY(220 days)	25.07	2.34	7.66	3.27	1.92	76.24

While this strategy effectively captures identifies statistical extremes, it exhibits reduced profitability during strong trending regimes. This is primarily because the volatility-weighted exit logic often triggers premature closures as the signal reaches extreme levels rapidly, causing the strategy to exit well before the trend concludes. Consequently, it fails to capture the full magnitude of profit available during sustained directional days.

3 Final Strategy

Intraday trading is modeled as a sequential decision-making problem in which an agent interacts with a market environment and learns a policy that maximizes expected cumulative reward. The setup is represented as a **Markov Decision Process (MDP)** defined by:

- **State (Observation):** At each timestep, the agent receives a fixed-length feature representation constructed from (i) a short rolling lookback window of market/technical features and (ii) current-step features, augmented with the agent’s internal state such as current position (flat/long/short) and mark-to-market (unrealized) PnL.
- **Action Space:** A compact discrete action space consisting of three actions: **Hold, Buy, and Sell**, which maps directly to trading decisions (maintain position, enter long/exit short, enter short/exit long).
- **Transition Dynamics:** At each step, open positions are updated based on price evolution and enforces exit conditions such as stop-loss, trailing-stop logic, and end-of-day liquidation.
- **Reward:** The reward function is shaped to align learning with risk-managed profitability. It incorporates rewards for profitable trades and penalties for undesirable behavior like excessive trading and adverse exits.

3.1 Rationale for Our Current Architecture

- **Reinforcement Learning** is selected because the trading objective is not a single-step prediction task but the optimization of a sequence of decisions under constraints. Trading performance is path-dependent: entry timing, holding behavior, and exit timing interact with transaction costs and risk controls.
- **Proximal Policy Optimization (PPO)** is selected for policy optimization due to its stability. PPO utilizes parallel rollout collection to gather diverse trajectories, ensuring the resulting training batch is diversified and less correlated.
- **An MLP policy** is used as it is well matched to tabular/engineered inputs and can learn nonlinear feature interactions without unnecessarily complex architectures.
- **The Actor-Critic architecture** is used to improve stability and support smoother convergence by training two distinct networks: the **Actor**, which determines the policy $\pi(a|s)$, and the **Critic**, which estimates the state value $V(s)$.

3.2 RL Hyperparameters and Parameterization

Hyperparameters are selected to balance exploration, stability, and learning speed in an intra-day environment.

3.2.1 Hyperparameter Tuning for Training Stability

- **Architecture:** Separate feed-forward networks are used for both the actor and critic, each with two hidden layers.
- **Rollout and Batching:** The rollout horizon n_{steps} is chosen to capture significant market structure, while clipping in PPO avoids destructive over-updates.

- **Discounting and Advantage:** The discount factor (γ) is set close to 1 to focus on long-term profitability. **Generalized Advantage Estimation (GAE)** with parameter λ_{gae} helps balance accuracy versus stability.

3.2.2 Optimization and Learning Rate Management

Observation and reward normalization are applied to stabilize gradient scales. To improve convergence, the learning rate is updated according to a linear schedule:

$$lr(t) = \frac{lr_{max} - lr_{min}}{T} \cdot t \quad (10)$$

Where $lr_{min} = 10^{-4}$, $lr_{max} = 5 \times 10^{-3}$, and $T = 1,000,000$.

3.3 State Space

The state space serves as the agent’s perceptual interface to the market environment, and its design fundamentally constrains the complexity of policies that can be learned and the efficiency with which learning occurs. In the context of algorithmic trading, the state space must capture market microstructure including price dynamics, volatility characteristics, momentum signals, and broader regime indicators. The state space is constructed as follows:

$$\mathbf{s}^{(t)} = \left[\mathbf{f}_{\text{lookback}}^{(t)}, \mathbf{f}_{\text{scalar}}^{(t)}, \mathbf{f}_{\text{agent}}^{(t)} \right]$$

where lookback features maintain a window of $w = 10$ historical timesteps, scalar features represent current values only, and agent features encode the current trading state. We organize our features into six distinct regime categories, each designed to capture specific aspects of market microstructure and dynamics.

1. **Price Action and Agent Features:** These features capture raw price movements and candle structure, providing the foundation for understanding immediate market dynamics. We compute logarithmic returns of the previous candle’s Open, High, Low, and Close prices. We also included the agent’s internal state, namely the position $\in \{1, 0, -1\}$ and mark-to-market PnL to make position-aware decisions.

$$r^{(t)} = \ln \left(\frac{P^{(t-1)}}{P^{(t-2)}} \right), \text{ where } P \text{ is the price}$$

2. **Momentum and Oscillator Features:** Momentum indicators such as RSI, CMO and CCI calculated over the lookback periods n (candles) $\in \{3, 5, 10, 20\}$, measure the velocity of price changes across multiple time-frames, helping identify overbought/oversold conditions, identify divergences between short and long-term momentum, and anticipate mean reversion opportunities. RSI values above 70 indicate overbought conditions; below 30 indicates oversold. CCI identifies cyclical turns and extreme prices. CMO ranges from -100 to +100, providing a centered momentum measure that distinguishes between trending and mean-reverting regimes.

$$RSI_n = 100 - \frac{100}{1 + \frac{\text{AvgGain}_n}{\text{AvgLoss}_n + \epsilon}}, \quad CCI_n = \frac{P^{(t)} - \text{SMA}_n(P)}{0.015 \times \text{MAD}_n(P)}$$

$$CMO_n = 100 \times \frac{\text{Gains}_n - \text{Losses}_n}{\text{Gains}_n + \text{Losses}_n + \epsilon}$$

3. **Volatility Features:** Volatility features quantify market uncertainty and risk, crucial for position sizing intuition and regime detection. Features like ATR and Rolling standard deviation normalized by price enable the agent to distinguish between low-volatility consolidation periods (where breakouts are likely) and high-volatility trending periods (where momentum strategies excel).

$$\text{ATR}_n = \frac{1}{n} \sum_{i=1}^n \text{TR}^{(t-i)}, \quad \text{ATR}_{\text{norm}} = \frac{\text{ATR}_n}{P^{(t-1)} + \epsilon}$$

$$\sigma_n = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (P^{(t-i)} - \bar{P}_n)^2}}{P^{(t-1)} + \epsilon}$$

4. **Trend Identification Features:** These features identify the prevailing market direction and trend strength, essential for aligning trades with dominant market forces. ER value above 0.3 indicates a trending market:

$$\text{ER}_{\text{binary}} = [\text{ER}_{20} > 0.3] \ \& \ \text{CHOP}_{\text{binary}} = [\text{CHOP}_{14} < 43.2]$$

Aroon crossovers signal trend changes – extreme values of Aroon up/down indicate strong directional conviction.

$$\text{Aroon}_{\text{down}} = \frac{n - \text{periods since } L_{\min}}{n} \times 100 \ \& \ \text{Aroon}_{\text{up}} = \frac{n - \text{periods since } H_{\max}}{n} \times 100$$

Alignment of regimes of all MAs indicates strong trends, while mixed regimes suggest transitions.

$$\text{Regime} = \begin{cases} 1 \text{ (Strong Bullish Trend)} & \text{if } \Delta\text{EMA} \geq 0 \text{ and } \text{EMA} > \text{EMA}_{\text{ref}} \\ 2 \text{ (De-accelerating Bullish Trend)} & \text{if } \Delta\text{EMA} < 0 \text{ and } \text{EMA} > \text{EMA}_{\text{ref}} \\ 3 \text{ (Strong Bearish Trend)} & \text{if } \Delta\text{EMA} \leq 0 \text{ and } \text{EMA} < \text{EMA}_{\text{ref}} \\ 4 \text{ (De-accelerating Bearish Trend)} & \text{if } \Delta\text{EMA} > 0 \text{ and } \text{EMA} < \text{EMA}_{\text{ref}} \end{cases}$$

5. **Temporal Features:** Intraday patterns exhibit significant time-of-day effects. We encode time using cyclical transformations:

$$H_{\sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad H_{\cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$

$$M_{\sin} = \sin\left(\frac{2\pi \cdot \text{minute}}{60}\right), \quad M_{\cos} = \cos\left(\frac{2\pi \cdot \text{minute}}{60}\right)$$

Cyclical encoding preserves the continuous nature of time and allows the agent to learn time-dependent patterns such as opening volatility and closing dynamics.

The state space design gives the agent a clear view of the market by using Multi-Scale Analysis to spot patterns over both short and long periods, while Regime Diversity ensures it recognizes different market types. To prevent confusion, Noise Reduction smoothes out random price jumps to reveal the true trend, and Normalization adjusts the numbers so the strategy works regardless of the asset’s price. The agent also uses Temporal Awareness to understand time-based patterns and Position Awareness to make smarter decisions based on what it currently owns and its current profit or loss.

3.4 Reward Function

The reward function is a critical component of our RL trading agent, as it directly shapes the agent's behavior and trading strategy. The following reward components are included in our reward function:

1. **Profit and Loss (PnL) Reward:** When the agent closes a position, it receives a reward proportional to the realized return expressed in basis points (bps). To instill loss aversion in the agent, we apply a higher penalty for loss making by multiplying the reward with a loss multiplier λ_{loss} .

$$\text{pnl}_{\text{bps}} = \left(\frac{P_{\text{exit}} - P_{\text{entry}}}{P_{\text{entry}}} \right) \times 10000 \quad \text{and} \quad r_{\text{pnl}} = \begin{cases} \text{pnl}_{\text{bps}} & \text{if } \text{pnl}_{\text{bps}} > 0 \\ \lambda_{\text{loss}} \cdot \text{pnl}_{\text{bps}} & \text{if } \text{pnl}_{\text{bps}} \leq 0 \end{cases}$$

2. **Stop loss hit penalty:** We implement a trailing stop mechanism to protect unrealized profits and provide a dynamic exit strategy. Let the price at time t be $p^{(t)}$, and define the trailing stop level as p_{trail} . The trailing stop mechanism is expressed as:

$$p_{\text{trail}} = \begin{cases} p_{\text{high}}^{(t)} \times (1 - \delta_{\text{trail}}) & \text{for long positions} \\ p_{\text{low}}^{(t)} \times (1 + \delta_{\text{trail}}) & \text{for short positions} \end{cases} \quad \text{where} \quad \begin{cases} p_{\text{high}}^{(t)} = \max(p_{\text{high}}^{(t-1)}, p^{(t)}) \\ p_{\text{low}}^{(t)} = \min(p_{\text{low}}^{(t-1)}, p^{(t)}) \end{cases}$$

We set the trailing stop loss level at $\delta_{\text{trail}} = 5$ bps and the static stop loss level at $\delta_{\text{stop}} = -5$ bps. When the trailing stop is triggered, the position is automatically closed with penalty $r_{\text{trail}} = \beta_{\text{trail}}$. When the static stop loss is triggered we impose a larger penalty given by $r_{\text{stop}} = \beta_{\text{stop}}$, to prevent catastrophic losses and discourage the agent from entering positions with high probability of immediate adverse movement.

3. **Missed Opportunity Penalty:** To encourage the agent to participate in significant market movements and avoid excessive passivity, we penalize remaining flat during volatile periods. We compute the absolute price movement over a lookback window $w = 5$ candles. If the price movement is more than the threshold $\theta_{\text{opp}} = 5$ bps we penalize for missing profitable moves, otherwise we give a small reward for not over-trading in calm markets.

$$r_{\text{opportunity}} = \begin{cases} \beta_{\text{miss}} & \text{if } \Delta_w > \theta_{\text{opp}} \\ \beta_{\text{wait}} & \text{if } \Delta_w \leq \theta_{\text{opp}} \end{cases} \quad \text{where} \quad \Delta_w = \frac{|p^{(t)} - p^{(t-w)}|}{p^{(t-w)}}$$

4. **Trade Entry Penalty:** To discourage excessive trading frequency and thereby excessive transaction fees, we also apply a fixed penalty upon entering any new position, given by $r_{\text{entry}} = \beta_{\text{entry}}$.
5. **End-of-Day Penalty:** This penalty encourages the agent to close trades before the trading session ends. The penalty is given by $r_{\text{eod}} = \beta_{\text{eod}}$.

The Table 3 summaries the rewards components. The complete reward at each timestep is the sum of all applicable components, included only when its corresponding triggering condition is satisfied at time t :

$$R^{(t)} = r_{\text{pnl}} + r_{\text{trail}} + r_{\text{stop}} + r_{\text{opportunity}} + r_{\text{entry}} + r_{\text{eod}}$$

Table 3: Reward Function Components and Parameters

Component	Parameter	Value	Purpose
Profitable Exit	—	$+r_{\text{pnl}}$	Reward realized gains
Losing Exit	λ_{loss}	$4.0 \times r_{\text{pnl}}$	Penalize losses asymmetrically
Stop Loss Hit	β_{stop}	-100	Severe penalty for poor entries
Trailing Stop Hit	β_{trail}	-10	Mild penalty for mechanical exits
Trade Entry	β_{entry}	-5	Discourage overtrading
Missed Opportunity	β_{miss}	-2	Penalize inaction in volatile markets
Wait Bonus	β_{wait}	+0.1	Reward patience in calm markets
End-of-Day Hold	β_{eod}	-10	Discourage overnight positions

Table 4: Average Annualized Performance Summary

Ticker	Ann.PnL(%)	Sharpe	Calmar	Max DD	Avg.Trades/day	Win Rate
EBX(255 days)	81.61	7.30	70.96	1.15	13.77	69.33
EBY(140 days)	77.97	4.91	63.91	1.22	18.85	69.08

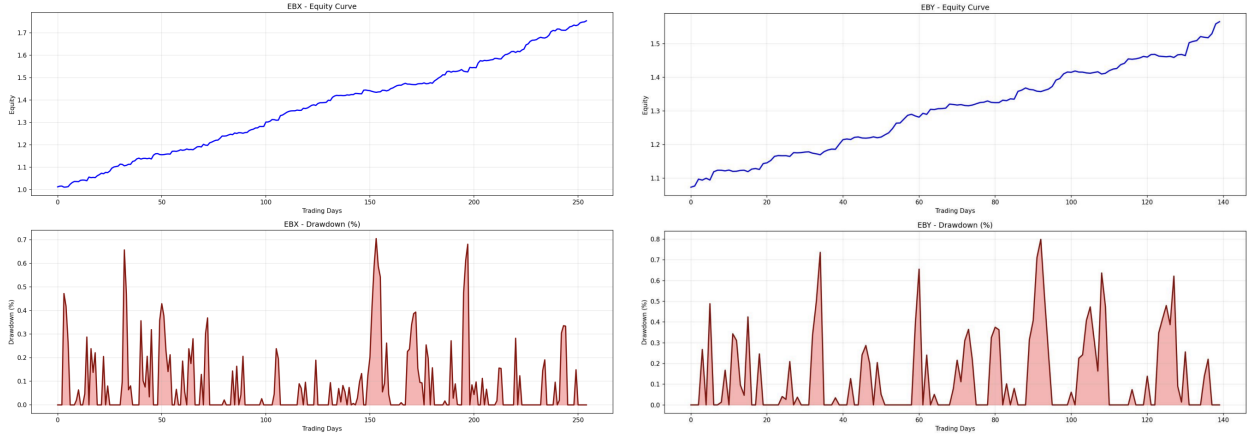


Figure 3: EBX Equity curve and EBY equity curve